

PROFILE

Mechanical engineer and industrial designer with 10+ years of experience developing hardware from concept through prototyping, validation, production release and documentation. Strong background in professional audio equipment, LiDAR/camera hardware, RF products, IoT sensors and industrial mechanisms. Focused on straightforward, intuitive mechanical solutions that can be manufactured, assembled and tested reliably.

PROFESSIONAL EXPERIENCE

Senior Mechanical Engineer | Bubblebee Industries

MAR 2025 - JUL 2026

Professional audio hardware development focused on microphone handling-noise reduction, robust product mechanics and validation-driven iteration.

- Design quick-release mechanism concepts and mechanical hardware for microphone blimp systems.
- Develop handling-noise reduction equipment for professional recording workflows.
- Build ESP32-based validation jigs with acceleration measurement and data logging.
- Use vibration and transmittance analysis to compare mechanical configurations and guide design decisions.

Senior Mechanical Engineer | Giraffe360 Ltd.

OCT 2023 - FEB 2025

Mechanical R&D and production support for a LiDAR-integrated camera hardware platform.

- Developed LiDAR-integrated brushless motor assemblies and injection-moulded components.
- Designed optical alignment and component-gluing tooling, adjustment frames and PCB test jigs.
- Validated designs using FDM, SLA, SLS, CNC, CFD and real assembly testing.
- Investigated assembly issues and worked with suppliers to improve manufacturability and process reliability.

Mechanical Designer | Warp Agency / Gravity Team Lustra

SUMMER 2023

Mechanical design and prototyping for a 2 x 7 m kinetic robotic chandelier.

- Created rail, pulley, actuator and support architecture for quiet, reliable movement.
- Built and validated workshop prototypes before ceiling installation.
- Prepared SolidWorks assemblies, part models and installation-ready mechanical details.

Mechanical Engineer / Industrial Designer | Riga Technical University, Science and Innovation Centre

SEP 2022 - OCT 2023

Mechanical engineering, industrial design and prototyping support for hardware startups and scientific projects.

- Developed sealed sensor assemblies, industrial mechanisms and custom prototypes.
- Designed a scaffolding load-cell sensor with PU overmoulding, LoRa antenna, wireless charging, sheet-metal covers and IP68 protection.
- Performed mechanical layout, load calculations, FEA support and production drawing preparation.

PROFESSIONAL EXPERIENCE - CONTINUED

Lead Mechanical Engineer / Industrial Designer | SAFtehnika JSC

SEP 2015 - SEP 2022

Led mechanical and industrial design for microwave radio equipment, IoT sensors and safety-system concepts from early sketches to mass production.

- Owned product architecture, prototypes, mechanical drawings, assembly manuals, BOMs, quality control and PDM implementation.
- Delivered Spectrum Compact, Integra-E2, the Aranet sensor family and fire-alarm product concepts.
- Applied DFM across injection moulding, CNC milling and turning, sheet metal, aluminium extrusion, 3D printing and silicone moulding.
- Designed IP54/IP68 product solutions and supported ATEX-certified equipment development.

Industrial Designer / Project Manager | Associates, Partners et Sons

2015

Industrial design and project delivery for bespoke design objects and consumer-product concepts.

- Created the Latvian Architecture Award object - a Rubik's-cube pineapple assembled from more than 50 CNC-milled parts.
- Developed a magnetic watch-strap concept with replaceable magnetic straps.

SELECTED PRODUCT HIGHLIGHTS

Aranet4 CO2 Sensor | *High-volume IoT product*

- Developed as part of the Aranet product family from initial concept through industrialisation.
- Produced in more than 100,000 units and recognised as Latvia's most innovative product of 2019.
- Used extensive prototyping to refine a semi-transparent enclosure suitable for varied interiors.

aranet.com/

Spectrum Compact | *0.3-80 GHz portable spectrum analyser*

- Lead engineer for a field product used by point-to-point radio installers and regulatory institutions.
- Combined CNC, sheet metal, silicone moulding, PCB packaging, IP54 design and complete production documentation.

www.spectrumcompact.com

Integra-E2 | *Point-to-point 5G radio equipment*

- Designed an aluminium-extrusion base and CNC-machined cover functioning as a heat sink.
- Delivered IP68-rated, ATEX-certified mechanical design, assembly definition and BOM documentation.

CORE SKILLS

Mechanical design: Product architecture, mechanism design, enclosure design, tolerance-aware CAD, assembly definition and design reviews

CAD and documentation: SolidWorks CAD, drafting and PDM; engineering drawings, assembly drawings, BOMs, manuals and production documentation

Manufacturing and DFM: Injection moulding, CNC milling/turning, sheet metal, aluminium extrusion, 3D printing, silicone moulding and supplier communication

Simulation and validation: FEA/CFD support, vibration/transmittance testing, load calculations, prototype validation and assembly-process testing

Test systems: ESP32 data logging, acceleration measurement, custom test jigs, PCB test fixtures and prototype-lab setup

Rendering: SolidWorks Visualize product rendering and technical presentation

EDUCATION

Automotive Engineering studies - Riga Technical University, 2008-2015

Professional bachelor-level studies; degree not completed.

REFERENCES

Janis Dobelis - Head of Engineering, Giraffe360

Maris Grinvalds - COO, SAFtehnika JSC

Additional references available on request.